

Stomatal conductance and transpiration of co-occurring seedlings with varying shade tolerance

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Abstract Forest ecosystems exert a large influence on the hydrologic cycle through transpiration, with the majority supplied by saplings and adult trees. However, a measurable amount is also supplied by seedlings. The contribution of seedlings is dependent upon species, which is ultimately controlled by microclimate. The objectives of this study were to (1) observe meteorological conditions of two forest microclimates; and (2) assess the intra- and interspecific stomatal conductance and transpiration responses of naturally occurring seedlings of varying shade tolerance. Naturally established seedlings in a deciduous forest understory and an adjacent clearing were monitored throughout the 2008 growing season in southeastern Pennsylvania (39°49'N, 75°43'W). The understory microclimate conditions overall had a lower degree of variability and had consistently lower mean quantum flux densities, air temperature, vapor pressure deficit, volumetric water content, and soil temperature than the clearing plot. Understory seedlings of *Fagus grandifolia* Ehrh. (American beech) and *Prunus serotina* L. (black cherry) had significantly lower mean monthly rates of water loss ($p = 0.05$) than clearing seedlings [*P. serotina* and *Liriodendron tulipifera* L. (yellow poplar)].

Additionally, water loss by shade-grown *P. serotina* seedlings was significantly lower ($p = 0.05$) than by sun-grown seedlings. Physiological differences, specifically shade tolerance, played an important role in determining the rates of stomatal conductance and transpiration in the seedlings. To a lesser degree, microclimate variability also influenced water loss. The results of this study validate results obtained in previous studies conducted largely under controlled conditions. Field validations are critical to developing better models and forest management strategies.

Keywords Transpiration · Seedlings · Stomatal conductance · Shade tolerance · Microclimate · Deciduous forest

Introduction

The study of transpirational water loss by trees in relation to microclimates is of considerable importance to forest management, specifically in terms of overall forest health and water supply availability. Transpiration (E) in forest ecosystems is controlled by the surrounding environment and physiological characteristics of individual species. Physical influences include photosynthetically active radiation (PAR), air temperature, vapor pressure deficit (VPD), wind speed, soil moisture, and soil temperature. Responses to these physical factors are ultimately controlled by stomatal conductance at the plant level according to physiological characteristics of the individual species (Willmer 1983). Shade and drought tolerance (Woods and Turner 1971; Chazdon 1988; Abrams et al. 1992; Sinclair et al. 2005), in addition to time of leaf emergence and development (Larcher 2002), are critical

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physiological traits determining E . Although water loss in plants is primarily controlled by the physiological characteristics of a species, the spatial distribution of a species is ultimately dictated by its microclimate. Daily, monthly and seasonal environmental and meteorological conditions in which shade-intolerant species grow are vastly different from those of shade-tolerant species, and therefore likely exert a large influence over the variability of water transport and water loss in each species and the surrounding microclimate. Although shade tolerance is a functional group characteristic, it can be argued that within a climatic region, all species descend from a common ancestor and share similar phylogenetic traits (Niinemets and Valladares 2006).

Accurate measurements of water loss in forest ecosystems are important for watershed and forest management, as forest understories may comprise as little as 5% up to 54% of the transpiration stream (Jones and Raynal 1986; Black et al. 1989; Baldocchi and Vogel 1996; Blanken et al. 1997). Strategies of water conservation in response to water and light availability among juvenile and mature trees are distinct, demonstrating the necessity of age-specific observations (Mediavilla and Escudero 2004; Piper and Cavieres 2009). Intra- and interspecific differences of E in saplings and mature trees have been well documented in the field and provide a clear picture of microclimate and physiological influences on water loss, including responses to radiation, humidity, air temperature, and soil water deficit (Mott and Parkhurst 1991; Granier et al. 2000; Bowden and Bauerle 2008; Herbst et al. 2008). In contrast, seedling E has traditionally been observed under manipulated settings, either in greenhouse conditions (Holmgren 2000; Nagakura et al. 2004; Baquedano and Castillo 2006) or in the field under an abbreviated range of microclimate variables (Wang et al. 2006). A controlled experimental setup provides a method of documenting specific microclimate influences and insures seedling survival, but fails to adequately demonstrate the entire array of meteorological forcings experienced by the seedling in reality.

The objectives of this study were to (1) observe meteorological conditions of two distinct microclimates and (2) assess the g_s and E responses of three seedling species of varying shade tolerances in their naturally occurring microclimates throughout an entire growing season with regard to overall specific characteristics and seasonal patterns. This study seeks to validate previous observations of seedling response to g_s and E as a result of variable light regimes and microclimates. The experimental design employed in this study insures that seedling responses are due directly to microclimate conditions or to physiological characteristics inherent to species growing in such microclimates and representative of actual circumstances.

Materials and methods

Study site and experimental design

Research was conducted at two study plots within the Bucktoe Creek Preserve in southeastern Pennsylvania (39.82°N, 75.72°W) throughout the growing season from 22 May until 6 October 2008. Annual summer daytime temperatures range from 13.4 to 24.7°C (30-year) with 104 mm of mean monthly precipitation (30-year) falling during the growing season (NCDC, 2000). The study site was located on well-drained, moderately eroded silt loam soils at a mean elevation of 106 m (PSU Soil Map, 2006).

A dense forest understory served as the first study plot, while an adjacent clearing served as the second plot (Table 1). Selection of species in this study was according to shade tolerance, an important functional trait that distinguishes species' response to light regimes and occasionally provides insight into numerous other environmental responses such as drought tolerance (Niinemets and Valladares 2006). Three species observed during this study in order of shade tolerance were *Fagus grandifolia* Ehrh. (American beech), *Prunus serotina* (black cherry), and *Liriodendron tulipifera* L. (yellow poplar) (Baker 1949). *Fagus grandifolia* seedlings were exclusive to the understory plot, *L. tulipifera* seedlings were exclusive to the clearing plot, and *P. serotina* seedlings were located in both study plots and therefore served as the control for microclimate and shade tolerance comparisons. *Fagus grandifolia* seedlings produce a shallow root system and are adapted to growing in very low light conditions with poorly drained soils (Tubbs and Houston 1990). *Prunus serotina* seedlings tolerate a range of soil moisture and light conditions, yet grow significantly faster in full light and produce a shallow and sprawling root system (Marquis 1990). *Liriodendron tulipifera* seedlings require high light levels for growth, moderately wet soils that are well-drained, and produce both deep and shallow roots (Beck 1990).

Twenty naturally established, 2nd-year seedlings were selected and tagged from each species group, for a total of 80 samples (*P. serotina* understory and clearing seedlings were considered as separate groups). Typically, shade-tolerant species have lower leaf mass per leaf area, lower stomatal density, and larger leaf size in mature adult trees (Valladares and Niinemets 2008). The number of leaves occurring per seedling on *F. grandifolia* and understory *P. serotina* individuals was lower than clearing *P. serotina* and *L. tulipifera* seedlings (Fig. 1) indicative of lower relative growth rates (RGR) and leaf area ratios (LAR = fraction of plant mass in leaves to dry mass per unit area) in shade-tolerant species as previously observed (Valladares and Niinemet 2008).

Table 1 Stand characteristics of the study site

	Understory plot	Clearing plot
Plot area (m ²)	2400	8000
Mean elevation (m)	102	96
Mean aspect (°)	193	56
Mean slope (°)	2.4	2.8
Maximum slope (°)	8.3	11.1
LAI (m ² m ⁻²)	4.52 ± 0.14	
Seedling density (ha ⁻¹)	7.26 × 10 ³ ^a 2.29 × 10 ⁴ ^b	2.55 × 10 ³ ^c 3.55 × 10 ⁴ ^d
Seedling height (m)	0.1–0.2 ^a 0.15–0.25 ^b	0.2–0.5 ^c 0.3–0.6 ^d
Leaves (seedling ⁻¹)	3–8 ^a 8–12 ^b	10–20 ^c 5–20 ^d

^a*F. grandifolia*; ^b*P. serotina* (understory); ^c*P. serotina* (clearing);
^d*L. tulipifera*

Porometry data

Porometry data were collected approximately weekly throughout the 2008 growing season to obtain measurements during the broadest range of microclimate conditions. Measurements began after full leaf emergence on 22 May to remove the influence of interspecific physiological differences on transpiration and stomatal conduction from leaf emergence. Subsequent data were collected on 30 May, 6 June, 9 June, 1 July, 7 July, 15 July, 22 July, 25 July, 31 July, 18 August, 25 August, 4 September, 11 September, 17 September, 29 September, and 6 October. By 6 October, the understory species displayed very low rates, so this was chosen as the last day of measurement to remove interspecific physiological differences in water loss associated with leaf senescence at the end of the growing

Fig. 1 Ideal representative of each seedling species group of **a** understory *Fagus grandifolia* Ehrh, **b** understory *Prunus serotina* L., **c** clearing *Prunus serotina* L., and **d** clearing *Liriodendron tulipifera* L.

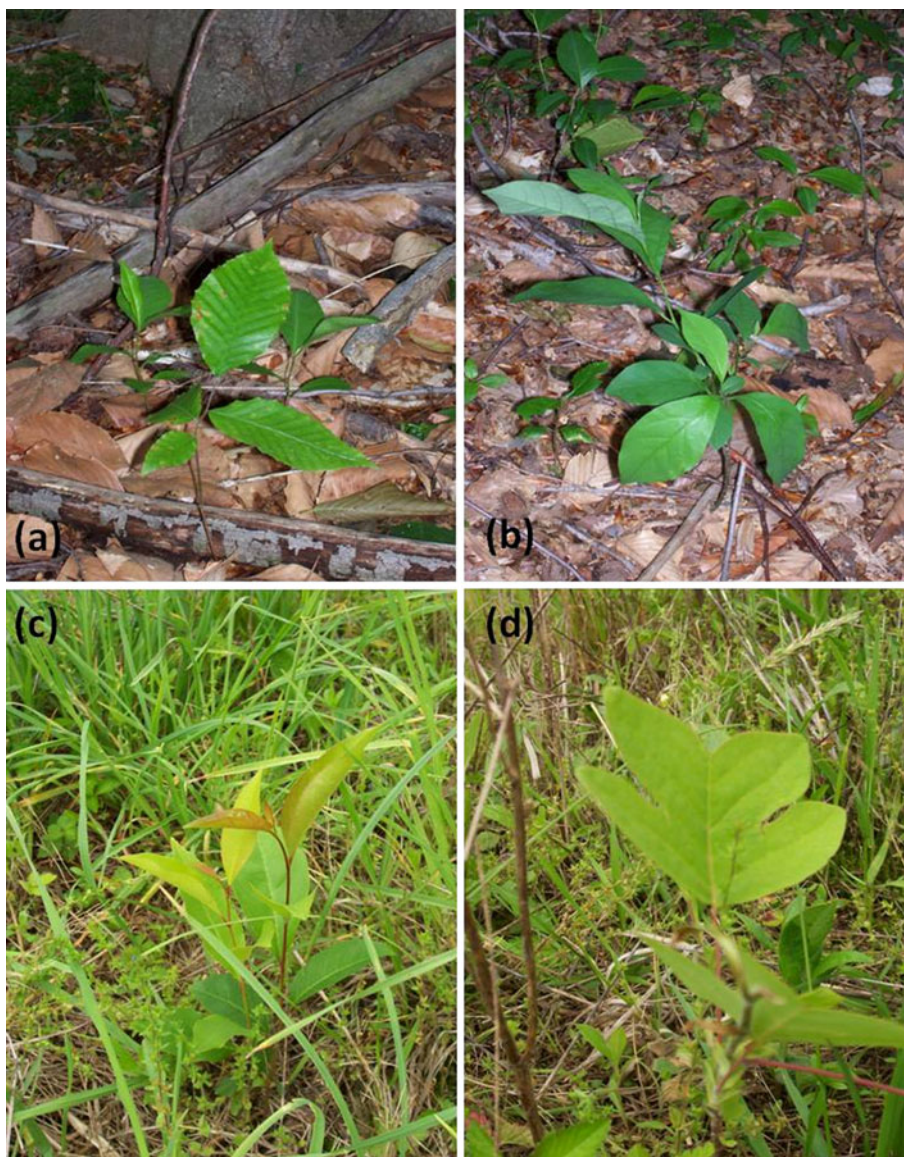


Table 2 Wilcoxon results of monthly microclimate conditions, where AT is air temperature, VPD is vapor pressure deficit, PAR is photosynthetically active radiation, ST is soil temperature, and VWC is volumetric water content

		AT	VPD	PAR	ST	VWC
May	Z score	18.46	17.71	49.82	31.33	50.35
	Prob > Z	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*
Jun	Z score	−10.61	−12.71	−86.59	−55.68	−87.57
	Prob > Z	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*
Jul	Z score	12.07	12.12	87.17	84.76	89.17
	Prob > Z	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*
Aug	Z score	−12.84	−10.84	−83.75	−76.08	−85.44
	Prob > Z	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*
Sep	Z score	−8.08	−10.05	−75.88	−47.94	−73.78
	Prob > Z	<0.0001*	0.0000*	0.0000*	0.0000*	0.0000*
Oct	Z score	−8.06	−4.15	−32.27	−31.87	−32.62
	Prob > Z	<0.0001*	<0.0001*	0.0000*	0.0000*	0.0000*

* Denotes a statistically significant p value

season. Recognizing the inherent physiological differences of stomatal conductance and transpiration during leaf emergence and senescence among species, this study collected porometry data after full leaf emergence until signs of senescence. Additionally, the porometer was not available from 9 June to 1 July for maintenance, so no data were collected during this 3-week period. Data collection times were scattered throughout the daytime to capture diurnal variability of meteorological conditions including temperature, humidity, and radiation. Times and dates of measurements are random because porometry data cannot be taken during periods of precipitation or when leaves are wet (i.e., immediately following precipitation, and early morning and late evening when dew is present).

Ten seedlings from each species group (*F. grandifolia*, understory *P. serotina*, clearing *P. serotina*, and *L. tulipifera*) were randomly sampled for a total of 40 sampled seedlings on each date of measurement. Total leaf conductance ($\text{mmol m}^{-2} \text{s}^{-1}$) and transpiration ($\text{mmol m}^{-2} \text{s}^{-1}$) were collected from a single leaf on each sampled seedling with a Li-Cor 1600 M Steady State Porometer (LI-COR, Lincoln, Nebraska). Values commonly found in the literature are given as stomatal conductance, so total leaf conductance was converted into stomatal conductance through the equation

$$g_s = (g_b g_l) / (g_b - g_l) \quad (1)$$

where g_l is the total leaf conductance as measured by the porometer and g_b is a known boundary layer resistance (0.15 s cm^{-1}) or conductance (6.7 cm s^{-1}) (Li-Cor, Inc., 1989). Leaf temperature and quantum flux densities were also measured. Quantum flux densities obtained in this way were used for analysis because they represent instantaneous values taken at the time of seedling measurement compared

to quantum radiation from the meteorological stations recorded every 5 min as discussed below.

Meteorological data

Meteorological data were obtained from two automated stations (Campbell Scientific, Logan, Utah) recording at 5-min intervals. The forest microclimate station (39.820°N , 75.724°W) was located within the understory plot and the clearing microclimate station (39.820°N , 75.727°W) was located in an adjacent field 750 m east of the clearing plot. Both stations measured temperature, relative humidity, solar radiation, *PAR*, wind speed and direction, soil temperature, *VWC*, and barometric pressure. *VPD* was calculated as the difference between the saturation vapor pressure and the actual vapor pressure at a given temperature and a measured relative humidity (Bolton 1980). Hourly precipitation data were collected with a vibrating-wire rain gauge encircled by an alter-type wind (Geonor A/S, Oslo, Norway) located next to the clearing station.

Data analysis

At both study plots, air temperature, *VPD*, *PAR*, soil temperature, and *VWC* were analyzed during hours of daylight when g_s and E were measured. On a monthly basis, the data were not normally distributed and required a statistical comparison using the nonparametric Wilcoxon test.

Repeated measures analysis of variance (ANOVA) was used to test for inter- and intraspecific differences of g_s and E among the following groups: (1) all species, (2) understory species, (3) clearing species, and (4) *P. serotina* seedlings in both microclimates. The application of

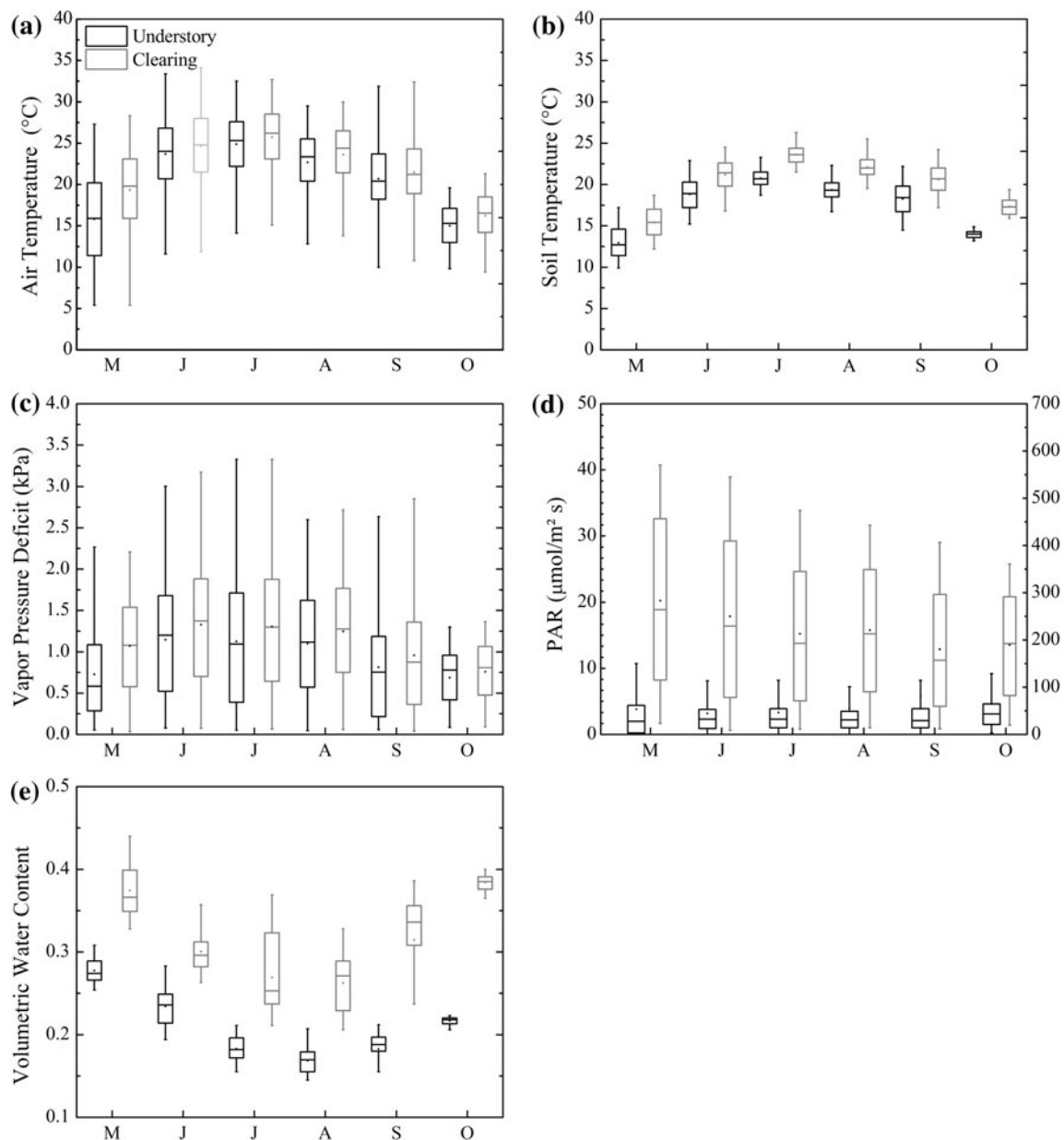


Fig. 2 Box plot summaries of microclimate conditions by month in the understory (solid outline) and the clearing (dotted outline) of **a** air temperature, **b** soil temperature, **c** vapor pressure deficit, **d** photosynthetically active radiation, and **e** volumetric water content. Boxes

represent one standard deviation, the horizontal bar within the box represents the median, the dot represents the mean, and the whiskers represent two standard deviations

repeated measures ANOVA accounts for duplicate measurements made on the same seedling throughout the study.

To further investigate the microclimate circumstances that influenced g_s and E , cluster analysis using Ward's method was performed. Principal components were selected from the meteorological variables (taken instantaneously at the time of seedling measurement) to isolate similar microclimate conditions experienced in the understory and the clearing and remove redundancy among the variables. Principal components were then introduced in the cluster analysis. The resulting clustered microclimate

occurrences were then compared to rates of water loss using MANOVA to determine the extent that microclimate played in regulating inter- and intraspecific g_s and E .

Results

Forest understory and clearing microclimates

Distinctions between understory and clearing microclimates were observed throughout the entire growing season.

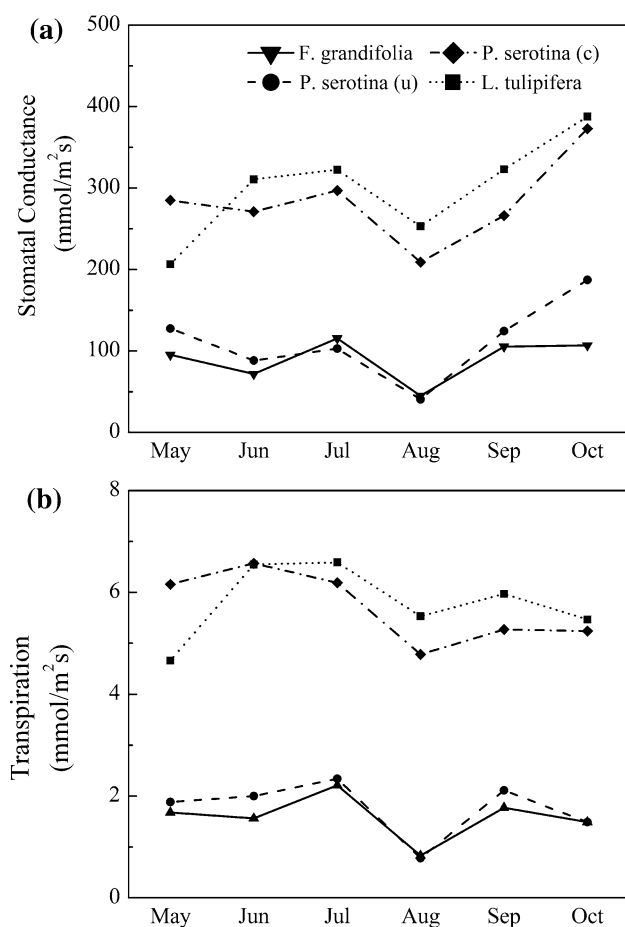


Fig. 3 Mean monthly course of **a** stomatal conductance and **b** transpiration for each of the four seedling groups

Air temperature, *VPD*, *PAR*, soil temperature, and *VWC* had significantly different means ($p = 0.0001$) between the understory and the clearing microclimates throughout all months of the growing season (Table 2). The understory plot had consistently lower mean air temperature, *VPD*, *PAR*, soil temperature, and *VWC* than the clearing plot (Fig. 2). Despite significantly different means, air temperature and *VPD* ranges were quite similar in both the understory and the clearing (Fig. 2a, c). The variability of *PAR* was a degree of magnitude greater in the clearing plot than in the understory (Fig. 2d), whereas soil temperature and *VWC* variability in the clearing were only moderately larger than in the understory (Fig. 2b, e).

Monthly trends in water loss

Seasonal trends of g_s and E observed at the monthly scale in the understory and the clearing are similar to contrasts observed between microclimates (Fig. 3). Shade-intolerant clearing species had consistently higher g_s and E compared to shade-tolerant understory species throughout the

growing season. *Fagus grandifolia* had the lowest mean monthly g_s and E , followed by understory *P. serotina* throughout the growing season. Clearing *P. serotina* initially had higher g_s and E than *L. tulipifera*, but the rank of these two species switched after May and maintained their new rank for the remaining months. All species displayed sharp declines in g_s and E from July to August, in accordance with declining soil moisture content (Fig. 3). However, the response of understory seedlings to declining soil moisture was much less dramatic than seedlings in the clearing (Fig. 3).

Prunus serotina and *L. tulipifera* seedlings growing in the clearing exhibited a much larger degree of intra-month variability than *P. serotina* and *F. grandifolia* understory seedlings (Figs. 4, 5). For example, monthly mean observations of g_s for understory *P. serotina* were $102.2 (\pm 72.9) \mu\text{mol m}^{-2} \text{s}^{-1}$ in May, $88.3 (\pm 48.5) \mu\text{mol m}^{-2} \text{s}^{-1}$ in June, $107.1 (\pm 56.9) \mu\text{mol m}^{-2} \text{s}^{-1}$ in July, $40.5 (\pm 25.5) \mu\text{mol m}^{-2} \text{s}^{-1}$ in August, $124.4 (\pm 45.2) \mu\text{mol m}^{-2} \text{s}^{-1}$ in September, and $90.7 (\pm 34.6) \mu\text{mol m}^{-2} \text{s}^{-1}$ in October (Fig. 4). Mean monthly g_s are similar for both understory species and display a moderate degree of variability as evidenced by the monthly standard deviations (Fig. 4). In contrast, shade-intolerant species growing in the clearing study plot had much larger monthly mean rates of g_s and larger standard deviations throughout all months of the growing season. Mean monthly g_s for clearing *P. serotina* for the same months were $284.8 (\pm 100.6) \mu\text{mol m}^{-2} \text{s}^{-1}$, $270.8 (\pm 101.9) \mu\text{mol m}^{-2} \text{s}^{-1}$, $281.2 (\pm 146.1) \mu\text{mol m}^{-2} \text{s}^{-1}$, $208.9 (\pm 96.2) \mu\text{mol m}^{-2} \text{s}^{-1}$, $266.1 (\pm 102.9) \mu\text{mol m}^{-2} \text{s}^{-1}$, and $372.8 (\pm 53.8) \mu\text{mol m}^{-2} \text{s}^{-1}$ (Fig. 4). Standard deviations of g_s observed throughout all months for clearing species is generally twice as large as the understory species, signifying the greater degree of variability observed in water loss mechanisms for shade-intolerant species. Mean monthly E and associated standard deviations are similar to those observed for g_s . Mean E is lowest for the most shade-tolerant species, *F. grandifolia*, and increases through the seedling groups with decreasing shade tolerance (Fig. 5). Shade-intolerant species of the clearing site displayed a much larger degree of variability than their shade-tolerant counterparts in the understory (Fig. 5). The larger response variability of g_s and E from clearing species suggests an inverse relationship between shade tolerance and microclimate variability.

Inter- and intraspecific water loss

Analysis of variance using repeated measures was performed on the four seedling groups to determine whether mean monthly g_s and E were significantly different between and among seedling groups ($p = 0.05$). Differences in g_s and

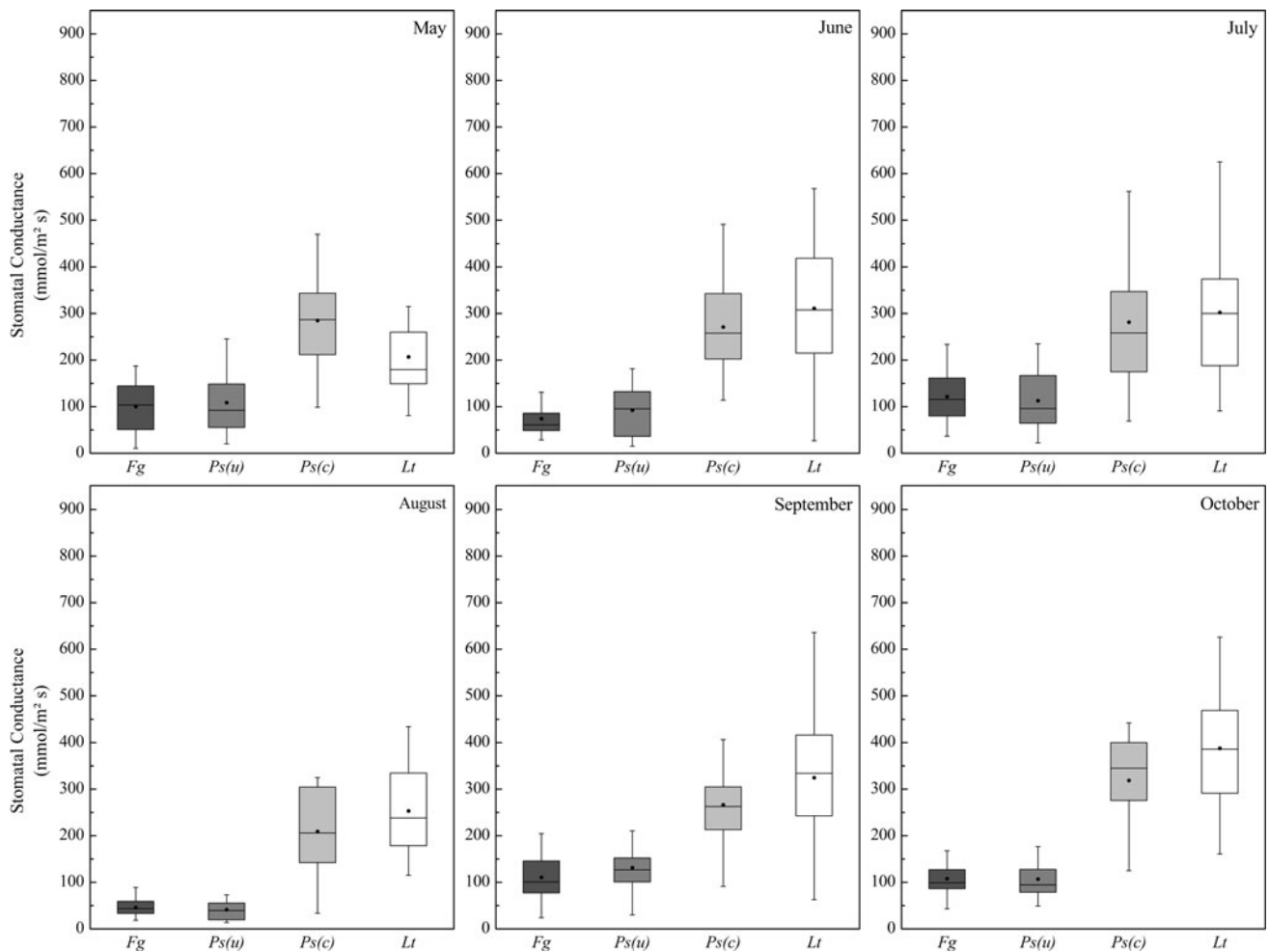


Fig. 4 Monthly box plots of mean g_s throughout the growing season of *F. grandifolia*, *P. serotina* (understory), *P. serotina* (clearing), and *L. tulipifera*, respectively. Boxes represent one standard deviation, the

horizontal bar within the box represents the median, the dot represents the mean, and the whiskers represent two standard deviations

E were significant in all months of the growing season when the four seedling groups were analyzed together (Fig. 6). The understory species, *F. grandifolia* and *P. serotina*, had statistically indistinct means throughout the growing season with the exception of September (Fig. 6). The clearing species, *P. serotina* and *L. tulipifera*, exhibited significantly different means in May and September, but were similar during the peak growing season (Fig. 6). In contrast, water loss between clearing and understory *P. serotina* seedlings was significantly different during all months of the growing season except May (Fig. 6).

Analysis of variance was also conducted on each seedling group to expose intraspecific variability of g_s and E throughout the growing season. When analyzed as four distinct seedling groups, interspecific and temporal g_s and E were observed ($p = 0.005$) with the combination of species and time also contributing to those differences (Table 3). The understory species did not produce significantly different g_s and E between species ($p = 0.291$), but

g_s and E for the understory seedlings in general were statistically different on a temporal scale (Table 3). The combination of factors, however, did not contribute to different g_s and E at the 95% confidence interval ($p = 0.083$) (Table 3). The clearing seedlings showed distinct g_s and E between species as well as temporally throughout the growing season, but the combination of factors once again was borderline insignificant at the 95% confidence interval ($p = 0.092$) (Table 3). Lastly, *P. serotina* seedlings exhibited distinct g_s and E both spatially by seedling microclimate and temporally, and through the combination of those two factors (Table 3). This result suggests that species living in more variable microclimates (i.e., the clearing plot) display more responsive g_s and E to changing environmental conditions where interspecific differences are strengthened and distinguishable (Figs. 4, 5). In contrast, under stable (i.e., the understory plot) and more limited (i.e., lower light regimes) conditions, interspecific differences are masked (Figs. 4, 5).

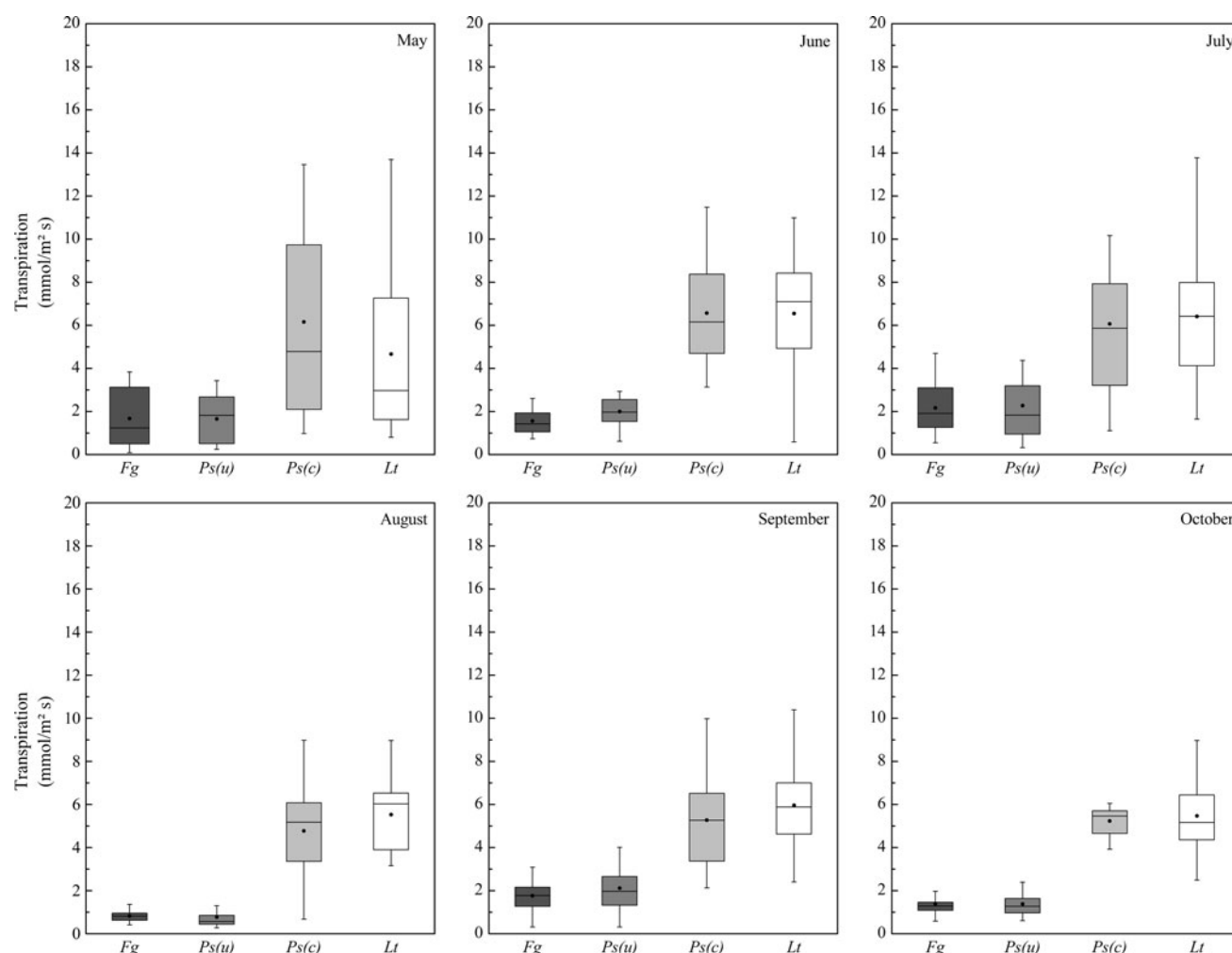


Fig. 5 Monthly box plots of mean E throughout the growing season of *F. grandifolia*, *P. serotina* (understory), *P. serotina* (clearing), and *L. tulipifera*, respectively. Boxes represent one standard deviation, the

horizontal bar within the box represents the median, the dot represents the mean, and the whiskers represent two standard deviations

Sun- and shade-grown seedling groups exhibited significantly different means throughout the growing season, with a significance level near zero (Fig. 6). *Prunus serotina* seedlings living in contrasting light regimes also displayed significantly different means throughout most of the growing season (Fig. 6). This suggests an environmental mechanism that is distinct from physiological differences between species (Figs. 4, 5). To expose the role of microclimate on g_s and E , clustering analysis was performed on microclimate conditions and analyzed against g_s and E of understory and clearing seedling groups. Microclimate conditions were converted into principal components to remove redundancy, producing four components for the understory (Table 4) and five components for the clearing (Table 5) with eigenvalues greater than one. Interpretation of the meaning of each component was obtained by analyzing the largest positive and negative loadings for each component. For example, the understory

PC1 has high positive loadings on air temperature, soil temperature, and leaf temperature indicating that this component functions as an overall temperature component and that these variables are positively related (Table 4). The understory components were interpreted to represent (1) temperature, (2) soil moisture availability, (3) vapor pressure deficit, and (4) sunflecks (Table 4). The clearing components were interpreted as (1) temperature, (2) vapor pressure deficit, (3) seasonal date, (4) time of day, and (5) solar radiation (Table 5).

Cluster analysis was performed separately on understory and clearing principal components under the assumption that the two microclimates were distinct and would inevitably produce separate clusters. Both microclimates produced five clusters, generally grouping together microclimate conditions temporally by time of day or season. The number of clusters for each microclimate was determined through both dendrogram analysis and

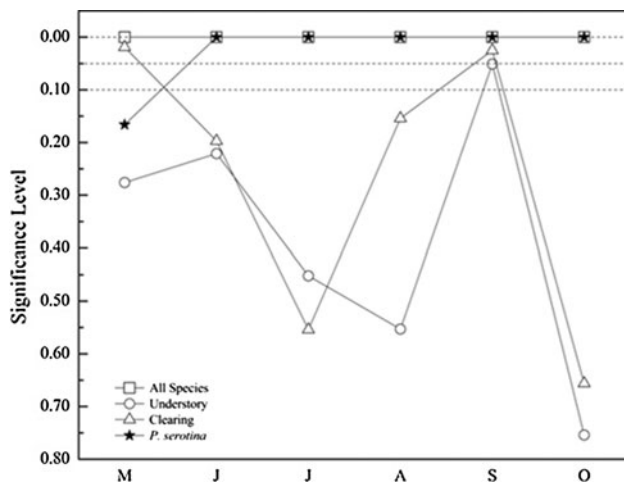


Fig. 6 ANOVA significance levels of mean monthly water loss for each of the four analyzed species groupings. Values above the lines of significance indicate statistically different g_s and E for that grouping within a given month. The grouping of all four seedling species was significant ($p = 0.000$) during all months

agglomeration coefficients in comparison to its corresponding clustering step. To determine how the clusters were created, loadings of principal components were plotted for each cluster. Individual clusters with large positive or negative principal component loadings are interpreted to be dominated by the microclimate conditions those principal components represent. The understory plot produced five clusters, which represented microclimates indicative of (1) the late spring, (2) peak summer, (3) times of intense sunflecks, (4) late summer, and (5) the first date of porometry measurements (Fig. 7). The clearing plot also produced five clusters which were determined to isolate microclimate conditions typical of the (1) late spring, (2) peak summer, (3) afternoon, (4) morning, and (5) late summer (Fig. 8). The five clusters for each plot are exemplary of the degree of microclimate variability experienced at each site. For the understory, the microclimate remained fairly constant throughout the course of the day, but responded to seasonal changes. In comparison, the microclimate clusters of the clearing plot suggest that conditions are variable not only throughout the season but also throughout a single day.

Each clustering scheme was introduced into a multivariate analysis of variance test (MANOVA), with the assumption that seedling g_s and E are also responding to changing microclimates. Stomatal conductance and E were significantly different between clusters in both study plots demonstrating the control of microclimate on seedling water loss. Over 8% of g_s and 23% of E in clearing seedlings and 3.6% of g_s and 7.8% of E in understory seedlings can be attributed to microclimate conditions as expressed through the clusters, according to partial Eta-squared

Table 3 Intra- and interspecific ANOVA results of water loss throughout the growing season, where WL represents water loss controlled by g_s and E , S signifies species, and T signifies time

	SS	df	MS	F	P value
All species ($n = 647$)					
Intercept	1.57×10^7	1	1.57×10^7	2627.6	0.000*
WL $\times$ S	4.16×10^6	3	1.39×10^6	243.94	0.000*
WL $\times$ T	9.62×10^5	16	6.01×10^4	10.580	0.000*
WL $\times$ S $\times$ T	1.06×10^6	47	2.25×10^4	3.965	0.000*
Clearing ($n = 324$)					
Intercept	1.86×10^7	1	1.86×10^7	2027.1	0.000*
WL $\times$ S	1.30×10^5	1	1.30×10^5	14.968	0.000*
WL $\times$ T	1.61×10^6	16	1.01×10^5	11.573	0.000*
WL $\times$ S $\times$ T	2.11×10^5	16	1.32×10^4	1.519	0.092
Understory ($n = 323$)					
Intercept	1.79×10^6	1	1.79×10^6	695.8	0.000*
WL $\times$ S	2.79×10^3	1	2.79×10^3	1.121	0.291
WL $\times$ T	2.23×10^5	16	1.39×10^4	5.601	0.000*
WL $\times$ S $\times$ T	6.16×10^4	16	3.85×10^3	1.549	0.083
<i>P. serotina</i> ($n = 322$)					
Intercept	6.55×10^6	1	6.55×10^6	980.5	0.000*
WL $\times$ S	1.49×10^6	1	1.49×10^6	223.9	0.000*
WL $\times$ T	4.85×10^5	16	3.03×10^4	4.537	0.000*
WL $\times$ S $\times$ T	4.93×10^5	16	3.08×10^4	4.615	0.000*

WL values are from specific dates of measurement, with approximately 20 observations for each species for each measurement date. T is analyzed as a nominal variable, one through six, for each month of the growing season

* Denotes a statistically significant p value

Table 4 Component matrix for microclimate principal components of the understory

	PC1	PC2	PC3	PC4
Julian date	−0.233	0.803*	−0.311	0.111
Air temperature	0.981*	−0.037	0.019	−0.086
Relative humidity	−0.573	0.342	0.665*	0.039
Wind speed	0.474	−0.553	0.120	0.075
Air pressure	−0.132	0.371	−0.624*	0.210
Soil temperature	0.763*	0.362	0.411	0.007
VWC	−0.331	−0.882*	−0.075	−0.060
Time	0.444	0.254	−0.618*	−0.164
Leaf temperature	0.915*	0.130	0.127	0.008
Leaf relative humidity	−0.103	0.556	0.458	−0.168
Quantum radiation	0.158	−0.068	0.107	0.993*

* Denotes highly correlated variables of each component as evident by large loading factors

values from the MANOVA (Table 6). Therefore, the microclimates of each study site can be further categorized into new sub-microclimates (time of day and time of

Table 5 Component matrix for microclimate principal components of the clearing

	PC1	PC2	PC3	PC4	PC5
Julian date	−0.020	−0.261	0.835*	0.163	−0.215
Air temperature	0.952*	0.134	−0.106	−0.092	−0.029
Relative humidity	−0.481	0.608*	0.365	−0.256	0.355
Wind speed	−0.014	0.483	−0.554	0.550	0.040
Air pressure	0.095	−0.631*	0.483	0.307	0.240
Soil temperature	0.833*	0.216	0.344	−0.113	0.068
VWC	−0.855*	−0.266	−0.144	0.180	0.116
Time	0.360	0.059	0.088	0.858*	0.123
Leaf temperature	0.955*	0.051	−0.091	−0.061	0.004
Leaf relative humidity	−0.111	0.797*	0.453	0.099	0.267
Quantum radiation	0.270	−0.475	−0.251	−0.161	0.726*

* Denotes highly correlated variables of each component as evident by large loading factors

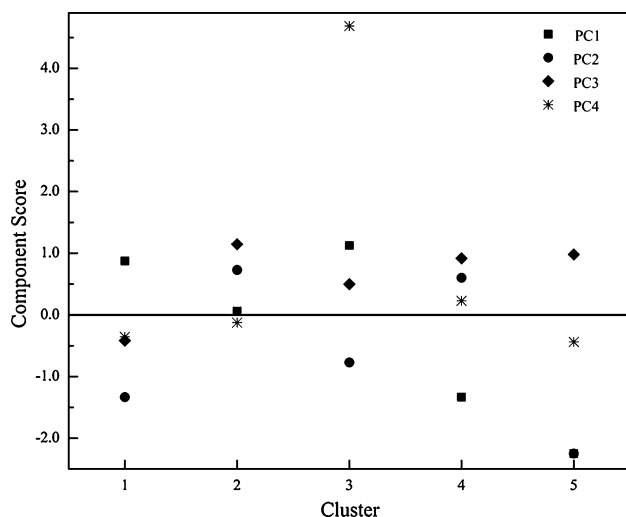


Fig. 7 Understory principal component score loadings based on cluster assignment. Scores closer to zero were considered neutral, more positive were positively related, and more negative were negatively related to the microclimate of each cluster. For example, cluster 1 has positive loadings for PC1, negative for PC2, and moderate to neutral for PC3 and PC4 suggesting a cluster of early- to mid-season measurements because of high air, soil, and leaf temperatures, low date, and high soil moisture with neutral influence from time of day and quantum radiation

season) that account for different rates of water loss for understory and clearing species. Additionally, E experienced more variation due to microclimate clusters than g_s , as revealed through partial η^2 values in Table 6. Transpiration, much like evaporation, is a process dominated by environmental influences, whereas g_s is controlled within the plant, so this result validates the method of analysis.

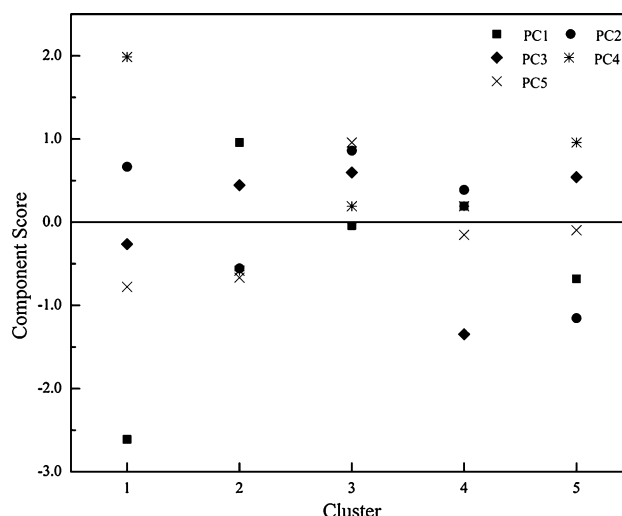


Fig. 8 Clearing principal component score loadings based on cluster assignment. Scores closer to zero were considered neutral, more positive were positively related, and more negative were negatively related to the microclimate of each cluster. For example, cluster 3 has moderate positive loadings for PC2, PC3, and PC5 and neutral loadings for PC1 and PC4 suggesting a cluster of afternoon measurements because of moderate humidity, date, and quantum radiation and neutral temperatures, soil moisture and time of measurement

Discussion

Microclimates

Higher air temperature, soil temperature, and VPD values in the clearing are a direct result of the difference between radiation regimes of the two microclimates imposed by a forest canopy. General conditions in the clearing plot therefore exhibited more optimal conditions for evaporation and consequently higher E . Additionally, the microclimate of the clearing plot maintained higher values of soil moisture. Although atmospheric evaporative demand would be higher in the clearing from the aforementioned microclimate conditions, the vegetational demand is much less due to the absence of mature trees and a forest canopy. Hence, greater VWC compensates for the higher VPD in the clearing, enabling greater g_s and E for seedlings grown in the clearing than in the understory (Yang et al. 2007).

Monthly g_s and E

All species had similar water loss curves during the growing season with initially increasing g_s and E following leaf emergence, then decreasing rates in response to peak summer water stress, followed by a moderate recovery as the water stress eased, concluding with a steady decline before senescence (Fig. 3). The observable decline in g_s and E in all seedling groups from July to August (Fig. 3)

closely follows the declining trend and minimum VWC in August at both study plots (Fig. 2d) demonstrated in previous studies (Jose et al. 2003).

Shade tolerance appeared to be the defining physiological characteristic in determining rates of water loss in all four seedling groups. The least shade-tolerant species, *L. tulipifera*, had the highest rates of g_s and E , with the exception of the onset of the growing season, followed closely by clearing *P. serotina* seedlings. This change of rank is most likely due to the documented physiological differences of leaf emergence and development of each species (Federer 1976); however, the long-term rank of water loss among the clearing species is in accordance with shade tolerance. The understory seedlings, with significantly lower rates of g_s and E , also maintained rank according to shade tolerance, where the intermediately tolerant understory *P. serotina* seedlings consistently had higher mean monthly than the most shade-tolerant species, *F. grandifolia*. Microclimate radiation regimes are critical determinants of g_s and E response to water stress, where seedlings growing under high light levels are more adversely affected during periods of water stress than those growing under low light levels (Yang et al. 2007; Quero et al. 2006; Abrams et al. 1992). *Liriodendron tulipifera* and clearing *P. serotina* seedlings demonstrated a much greater decline in g_s and E in response to water-stressed conditions in July and August in contrast to much more subtle response of understory *P. serotina* and *F. grandifolia* seedlings (Fig. 3).

Inter- and intraspecific water loss

Seedlings grown in contrasting microclimates were documented to have statistically different g_s and E throughout the growing season as demonstrated through ANOVA results. Although *L. tulipifera* and *P. serotina* clearing seedlings had significantly different rates of water loss throughout most months of the growing season, the understory seedlings did not. This stronger sensitivity to changing microclimates by species with larger g_s and E has been previously documented (Oren and Pataki 2001). Figures 3 and 4 suggest that interspecific physiology (i.e., shade tolerance) contributes to g_s and E as illustrated between the clearing and understory species, but also an additional forcing may be responsible for some degree of the response as illustrated by significantly different responses of *P. serotina* growing in the understory and the clearing.

The existence of significant inter- and intraspecific differences exposed in this study implies environmental response of species beyond interspecific physiological traits. The greater stability of understory microclimates in comparison to clearing microclimates was evident in this

Table 6 MANOVA results of microclimate clusters compared to g_s and E

	SS	df	MS	F	P value	η_p^{2*}
Understory						
g_s	5.45×10^4	4	1.36×10^4	3.086	0.016*	0.036
E	3.88×10^1	4	9.71×10^1	7.040	0.000*	0.078
Clearing						
g_s	7.25×10^5	4	1.18×10^5	7.164	0.000*	0.082
E	5.76×10^2	4	1.44×10^2	24.070	0.000*	0.232

* Partial eta-squared values express how much of a variable is controlled by clustering out of 1

study, especially in terms of PAR and VWC. Daily, although not seasonal stability, persists in the understory as exposed through cluster analysis. Therefore, frequently changing radiation regimes and more variable water availability (and atmospheric demand) in the clearing induced a large range of responses from the shade-intolerant species, exposing and highlighting interspecific differences and responses to changing microclimates. In comparison, understory microclimate conditions were less variable, with changes occurring gradually and mitigated by the forest canopy. Consequently, understory *P. serotina* and *F. grandifolia* seedlings responded with much less variability than their clearing counterparts. Understory seedlings had near different means in September, a month during which temperatures were moderate and soil moisture was not limited, possibly maximizing interspecific differences. This conclusion was also demonstrated through the partial η^2 values from multivariate analysis of variance of the microclimate clusters (Table 6).

Conclusion

This study highlighted inter- and intraspecific differences that exist among co-occurring seedling species in dissimilar microclimates. Understory seedlings experienced more similar rates of g_s and E than clearing seedlings at the seasonal level and responded less dramatically to water stress conditions. This distinction was controlled primarily by physiological factors and secondarily by microclimate differences. The significance of this work lies in the ability to distinguish between rates of water loss through g_s and E among seedling species using shade tolerance as a proxy. Shade tolerance proved to be an adequate means to determine rank of g_s and E of species adapted to the same microclimate or between different ones.

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